

Kainoa (Kai) Nishida

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EDUCATION

Cornell Tech

M.Eng. in Computer Science

August 2026 - June 2027 (Expected)

GPA: 3.95

University of California, Irvine (UCI)

B.S. Computer Science — Minor in Mathematics

September 2022 - June 2026 (Expected)

GPA: 3.95

- **Coursework:** Algorithms, Data Structures, Databases, Machine Learning, Neural Networks, Probabilistic Learning, Graph Algorithms, Game Theory, Probability, Advanced Linear Algebra, Discrete Mathematics
- **Awards:** ICS Honors (2024), Summer Undergraduate Research Scholarship (2025), National Merit Finalist (2022), Phi Beta Kappa Book Award (2023), Dean's List (All Quarters), UC Regent's Scholar (All Quarters)

RELEVANT EXPERIENCE

Google

Software Engineer Intern

San Jose, CA

May 2026 – August 2026

- Incoming

Amazon

Software Development Engineer Intern

Bellevue, WA

June 2025 – September 2025

- Designed and built a Lambda-driven workflow automator for an internal change-management tool using an AWS Bedrock multi-agent system, backed by an S3-based rolling-buffer ingestion pipeline.
- Drove a ~80% reduction in high-severity incidents during beta; projected to save 72,000+ engineer-hours/year (~18 hours/request × ~4,000 requests) with metrics, alerts, and documentation owned end-to-end.

Zhang Laboratory

Machine Learning & Genomics Research Assistant

Irvine, CA

January 2025 – Present

- Packaged research code as a PyPI CLI (Tyro), enabling 50+ lab members to run standardized training/evaluation with minimal setup friction using Slurm scheduler.
- Prototype VAE-ODE pipelines for RNA velocity and run distillation experiments on DNA foundation models to accelerate inference while preserving accuracy.

Commit the Change

Software Engineer

Irvine, CA

November 2022 – June 2025

- Delivered production software for 3 nonprofits (Get Inspired, Feeding Pets of the Homeless, Colette's Children's Home), supporting 700+ staff and volunteers.
- Developed backend routes, database schema, and automated triggers in MySQL to streamline nonprofit logistics and reduce manual data entry.

Irvine Spatial Neuroscience Laboratory

Software Engineer

Irvine, CA

June 2024 – June 2025

- Led a migration from ShadCN to Chakra UI, unifying design tokens and theming, unlocking richer components, and enabling faster feature iterations with full parity.

LEADERSHIP

Commit the Change

Technical Lead and Education Director

Irvine, CA

August 2023 – June 2025

- Led a cross-functional team of 12 developers and 5 designers to take projects from requirements and design through launch and iteration.
- Ran professional development workshops for 30+ developers, covering industry practices and technical growth.
- Organized and facilitated 100+ mock interviews and mentoring sessions, providing individualized feedback and career guidance.

Donald Bren School of ICS

Learning Assistant, CS 161: Design & Analysis of Algorithms

Irvine, CA

September 2025 – December 2025

- Supported 206 students through weekly office hours and 1:1 guidance; clarified proofs, runtime and memory complexity, and algorithm design patterns.

PROJECTS

RNA Flux – Zhang Laboratory

PyTorch · NumPy · SciPy · AnnData · Tyro CLI · Slurm

- Built a **Bayesian RNA-velocity** model (VAE + ODE-constrained kinetics) to learn latent cell time/state and gene-specific transcription rates via variational inference.
- Recovered **continuous trajectories** and **fate bifurcations** with neural, differential-equation generative modeling across single-cell datasets.

Sustainably – Caltech Hackathon

React · Node.js/Express · Chakra UI · Python (ML) · REST APIs

- Built a search engine that surfaces clothing from eco-friendly companies by training a deep-learning model to score “eco-friendliness” from multiple signals.

Fabflix – University of California, Irvine

Java · Tomcat · JDBC · MySQL · AWS · Kubernetes · Apache HTTPD · jQuery · reCAPTCHA

- Developed a full-stack movie e-commerce platform on AWS with Kubernetes orchestration, MySQL replication, and Apache load balancing.
- Implemented secure login, cart checkout, full-text search with caching, and reCAPTCHA bot protection.

SKILLS

Languages & Frameworks: Python, C++, C, Java, JavaScript, TypeScript, HTML/CSS, Bash, React.js, Next.js, Node.js, Express.js, Django

Tools: Git/GitHub, Figma, Linux/Unix, Docker, DigitalOcean, Firebase, MongoDB, PostgreSQL, Kubernetes, AWS, Render, Splunk, ServiceNow

Data Science & Machine Learning: SQL, NumPy, Pandas, Seaborn, Matplotlib, scikit-learn, PyTorch